

Biological Databases

1. What are biological databases?

2. Primary databases

1. EMBL / ENA
2. DDBJ
3. GOLD
4. UniProt
5. RCSB PDB

3. Secondary databases

1. InterPro
2. OMIM
3. Gene Ontology
4. STRING

4. NCBI-Entrez

1. Protein
2. Gene
3. PubMed / PM Central
4. Nucleotide
5. GenBank

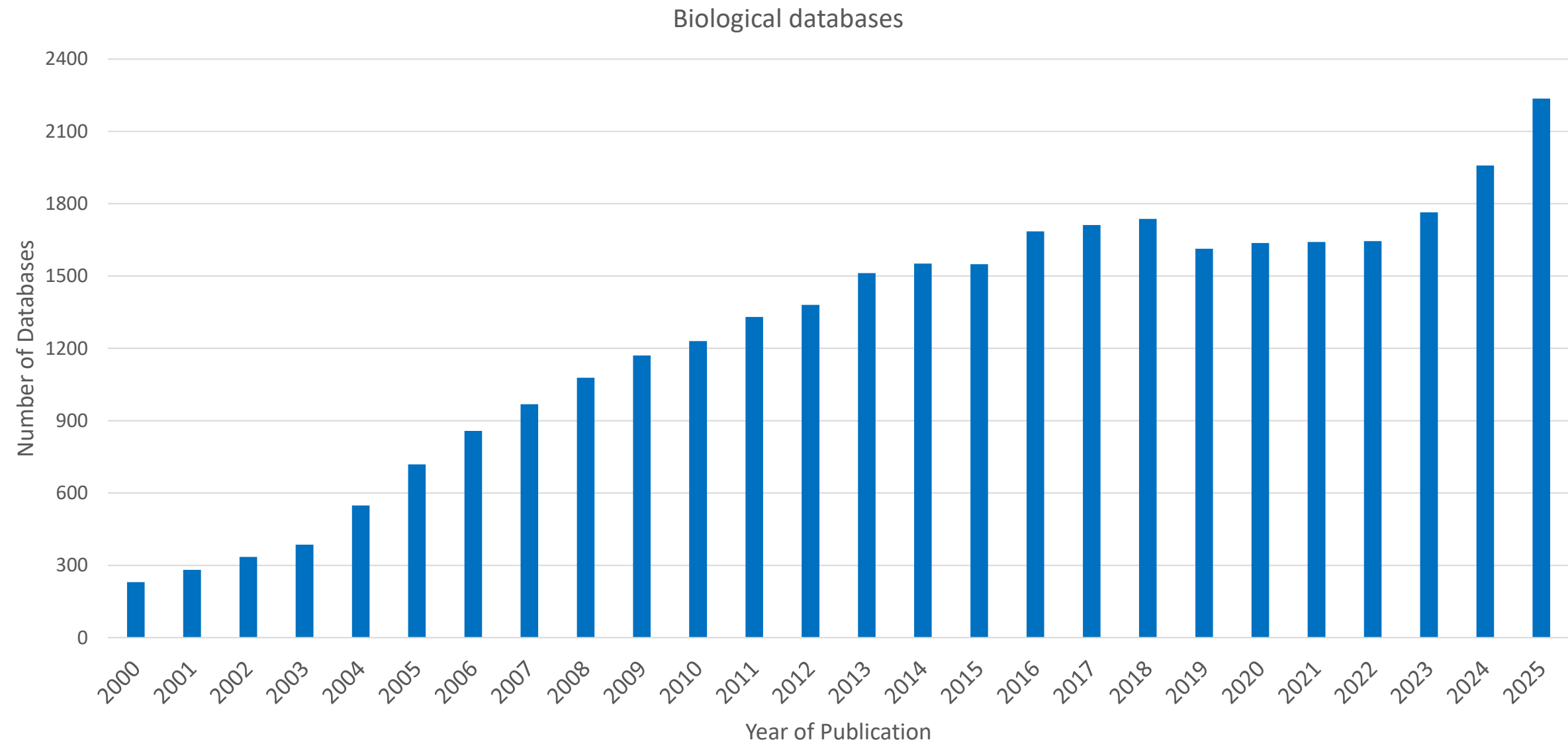
- In this course I will give you a small introduction to all these different databases, without going to much into detail about the general principles, algorithms and data derivations behind them
- Later on in your working life, when you need to work in-depth with one of these databases you should look up these things to get a deeper understanding of the methodology behind the databases entries and to be able to analyze the results you get in a better way

- Be aware that trying to understand all principles behind a database, especially a secondary database, can be a very tedious process
- A good example is the database STRING that has 13 publications about the database itself and you would need read through at least 30 to 50 additional publications to understand all the principles behind it and other databases it takes its data from

What are biological databases?

- Nearly all biological and biochemical data is collected in online databases and scientists should always try to publish their data in such databases to make it available to other scientists
- The most important ones are sequence databases which are the basis for searches with algorithms like FastA and BLAST as well as the structural databases like the RCSB-PDB or the AlphaFold database
- We need a big number of interconnected databases, to run studies in systems biology, that means to analyze cross links between nucleic acid data and protein data or between protein sequences and evolutionary data

- To handle these tasks many databases were developed over the years and their number is still growing to this day
- At the beginning of each year, starting in the year 2000, the Journal Nucleic Acid Research publishes a database Issue in which it updates its collection of biological relevant databases
- Currently were nearing a number of 2000 online available databases that deal with biological, biochemical, biophysical, medical and pharmaceutical data
- The catalogue is available online: <https://www.oxfordjournals.org/nar/database/c/>



- When talking about biological databases we have to differentiate between primary and secondary databases
- **primary databases** collect experimental data directly from scientists and examples are:
 - EMBL/ENA
 - Uni-Prot (hybrid)
 - RCSB-PDB
- **secondary databases** contain derived knowledge on the basis of the data from primary databases and examples are:
 - SMART
 - STRING
 - InterPro

- In general the difference between primary and secondary databases is very easy
- However over the course of time some databases emerged that are somewhat of „hybrid“ databases
- These databases collect data like primary databases and annotate it with the help of different tools that make use of known information just like secondary database do
- Therefore it's not always easy to determine the type of a database just by looking at the saved data

- A good example for such „hybrid“ databases is the UniProt
- The UniProt was originally created as a Primary database but nowadays makes us of it's Sister databases for annotations and interpretation of the uploaded sequences [39]
- Since it also contains protein sequences that are translated from uploaded nucleotide sequences they tread on a path directly between primary and secondary databases

Primary databases

- As mentioned earlier primary databases collect experimental data about certain topics directly from scientists and every scientist should upload the results of his or her experiments to these databases
- Here we will discuss some primary databases for working with nucleotide sequences, with protein sequences or with structural data of proteins
- Some primary databases have functional equivalents depending on the country they stem from
- The most classical example for this phenomenon are the EMBL / ENA [3], DDBJ [4] and GenBank [5] which are nucleotide archives for Europe, Japan or the USA respectively

- Having different databases hosted in Europe, America or Asia made sense in the beginning of these databases, since there were no big internet connections between Europe and The US for example
- Maintaining multiple databases that contain the exact same data, DDBJ, GenBank and ENA run daily updates to make sure all three contain the same data, seems a bit redundant in today's world
- It may be useful to have different web tools to access the data, but the data in itself should be saved on only one server and not three or more

- Stands for International Nucleotide Sequence Database Collaboration (INSDC)
- Is a long-standing foundational initiative that operates between DDBJ, EMBL-EBI and NCBI
- INSDC covers the spectrum of:
 - ▶ data raw reads (output from sequencing machines)
 - ▶ alignments (comparisons between multiple sequences)
 - ▶ assemblies (chromosomes, etc.)
 - ▶ functional annotation (coding regions, non-coding regions, etc.)
- Annotated with contextual information relating to samples (organisms, etc.) and experimental configurations (machine set up, extraction experiment, etc.)

Data type	DDBJ	EMBL-EBI	NCBI
Next Generation reads	Sequence Read Archive	European Nucleotide Archive	Sequence Read Archive
Assembled Sequences	DDBJ		GenBank
Samples	BioSample		BioSample
Studies	BioProject		BioProject

- Free and unrestricted access to all of the data records their databases contain
- Appropriate credit is given by citing the original submission
- The INSDC will not attach statements to records that restrict access to the data
- Specifically, no use restrictions or licensing
- All database records submitted to the INSDC will remain permanently accessible as part of the scientific record
- Submitters are advised that the information displayed on the Web sites maintained by the INSDC is fully disclosed to the public
- Only a limited editorial control and some internal integrity checks are done by the database the rest is the responsibility of the submitting author

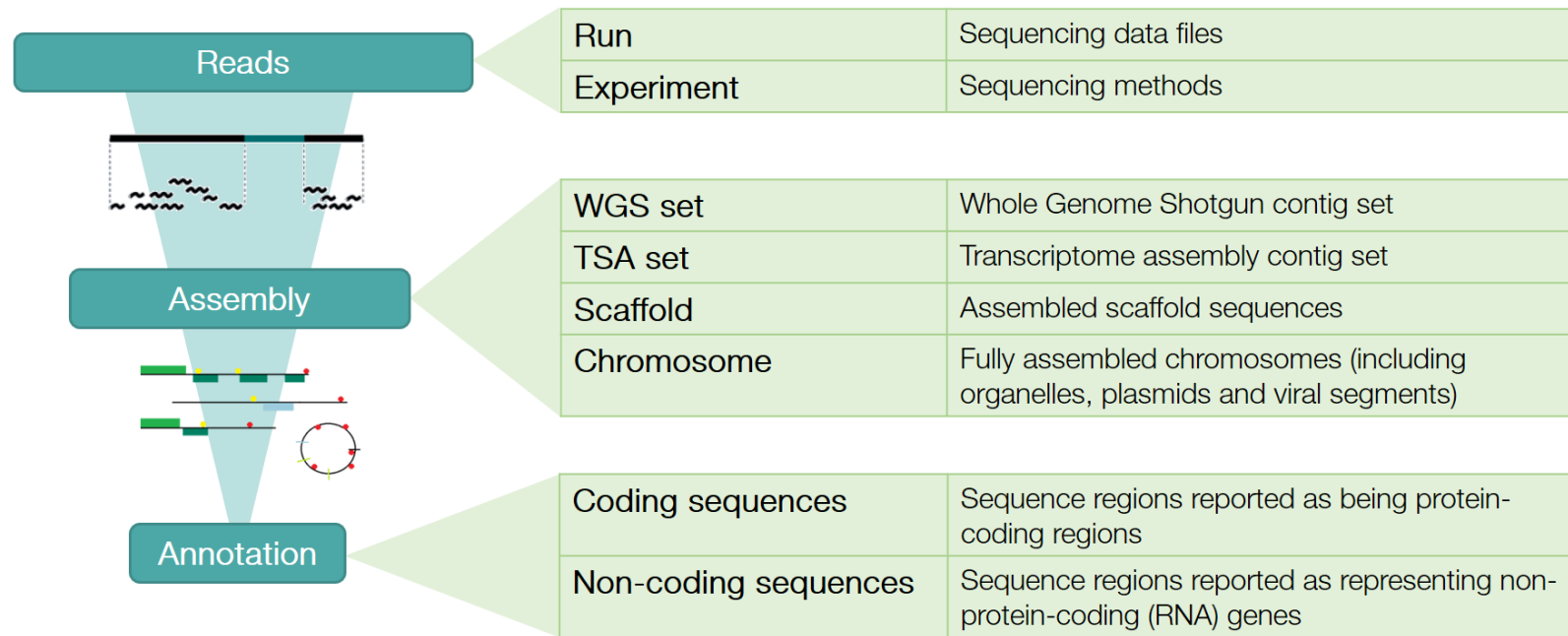
- The European Nucleotide Archive, comprising the EMBL Nucleotide Sequence Database and the Ensemble Trace Archive, is a collection of DNA sequences
- This database is available under the following link:
<https://www.ebi.ac.uk/ena/browser/home>
- Covers raw sequencing data, sequence assembly information, functional annotation as well as experimental data
- Presents all kind of data that resolves around nucleotide sequencing, including isolation, source material and so on



- THE ENA separates data into three categories
- input information:
 - ▶ sample
 - ▶ experimental setup
 - ▶ machine configuration
- Output machine data:
 - ▶ sequence traces
 - ▶ reads
 - ▶ quality scores
- Interpreted information:
 - ▶ assembly
 - ▶ mapping
 - ▶ functional annotation

- Data arrives at ENA from a variety of sources:
 - ▶ submissions of raw data
 - ▶ assembled sequences and annotation from small-scale sequencing efforts
 - ▶ data provision from the major European sequencing centres
 - ▶ routine and comprehensive exchange with its partners in the International Nucleotide Sequence Database Collaboration (INSDC)

- ENA is organised into three tiers: reads, assemblies and annotations



ENA Mainpage

ENA Example Sequence

ENA Example Coding

ENA Example Project

- Stands for DNA Data Bank of Japan and collects DNA sequences
- It is located at the National Institute of Genetics (NIG) in the Shizuoka prefecture of Japan and online available at: <https://www.ddbj.nig.ac.jp/index-e.html>
- Like the ENA it is part of the INSDC
- DDBJ began data bank activities in 1987 at NIG
- Remains the only nucleotide sequence data bank in Asia



- DDBJ Center is officially certified to collect nucleotide sequences from researchers
- Exchanges the released data with EMBL/ENA and NCBI on a daily basis, the three data centers share virtually the same data at any given time
- Collects sequence data mainly from Japanese researchers, but of course accepts data and issue the accession numbers to researchers in any other countries
- Provides software tools for data analyses developed by DDBJ
- Hosts a state-of-the-art NIG Supercomputer which allows external users to run their calculations

DDBJ Mainpage

DDBJ Biosample

DDBJ Project

DDBJ ARSA (Search Tool)

- Genomes On Line Database is a comprehensive resource for the centralized monitoring of genome and metagenome projects worldwide
- Both complete and ongoing projects, along with their associated metadata, can be accessed
- Goal of providing the most comprehensive repository of metadata information related to the projects and their organisms/environments



- The study of genetic material recovered directly from environmental or clinical samples by sequencing
- Traditional microbiology and microbial genome sequencing and genomics rely mostly upon cultivated clonal cultures
- Environmental gene sequencing clones specific genes to produce a profile of diversity in a natural sample
- It revealed that the vast majority of microbial biodiversity had been missed by cultivation-based methods

- The GOLD database is available under <https://gold.jgi.doe.gov/>
- Is based on a four-level project classification system containing:
 - ▶ Studies
 - ▶ Biosamples / Organisms
 - ▶ Sequencing Projects (SPs)
 - ▶ Analysis Projects (APs)

- Study is an umbrella project
- Broadly defines the overall goal of a research proposal and outlines the key objective of its underlying projects
- It contains a list of sequencing projects that are a part of the original study
- ‘Proposal’ is a synonym for ‘Study’ (e.g., HMP study, GEBA study, etc.)
- Simply said a Study is one research project

- Biosample is a description of the environment from where a DNA/RNA sample was collected
- Contains metadata such as
 - ▶ a habitat
 - ▶ an ecosystem
 - ▶ a geographical location
 - ▶ a latitude/longitude
 - ▶ etc.
- Currently, the GOLD Biosample entities are defined for all metagenome and metatranscriptome samples
- DNA and RNA extracted from the same physical sample can be used for a metagenome and a metatranscriptome project, respectively

- Organism is a description of an individual entity such as a bacterium, a fungus, a plant, an animal, or a virus
- Can be a cultured isolate of a pure strain of a bacterium or an uncultured single cell isolated using cell sorting
- All Organisms in GOLD are required to have basic taxonomic information such as a phylum, a genus, a species, a strain, and an NCBI taxonomy ID
- Defining an Organism entity is essential for GOLD Sequencing Projects with Whole Genome Sequencing (WGS) and Transcriptome sequencing strategies
- For more information about how organisms can be identified see [40 – 46] or the zip-file ABI_Paper_GOLD

- Sequencing Project is a description of the sequencing output from an individual Organism or Biosample
- An individual Sequencing Project may be composed of more than one sequencing reaction and/or sequencing technology
- A Sequencing Project may be for an isolate genome, a single cell, a metagenome, a metatranscriptome, etc.
- One sequencing project can thus contain more than a single sequence

- Analysis Project is a description of the informatics processing of a sequence associated with a corresponding Sequencing Project
- It specifies how the assembly and annotation of the sequence were performed
- Defining an Analysis Project entity is essential for submitting dataset to IMG (Integrated Microbial Genomes) for analysis
- An individual submission in IMG represents an individual Analysis Project in GOLD

GOLD Mainpage

GOLD Study

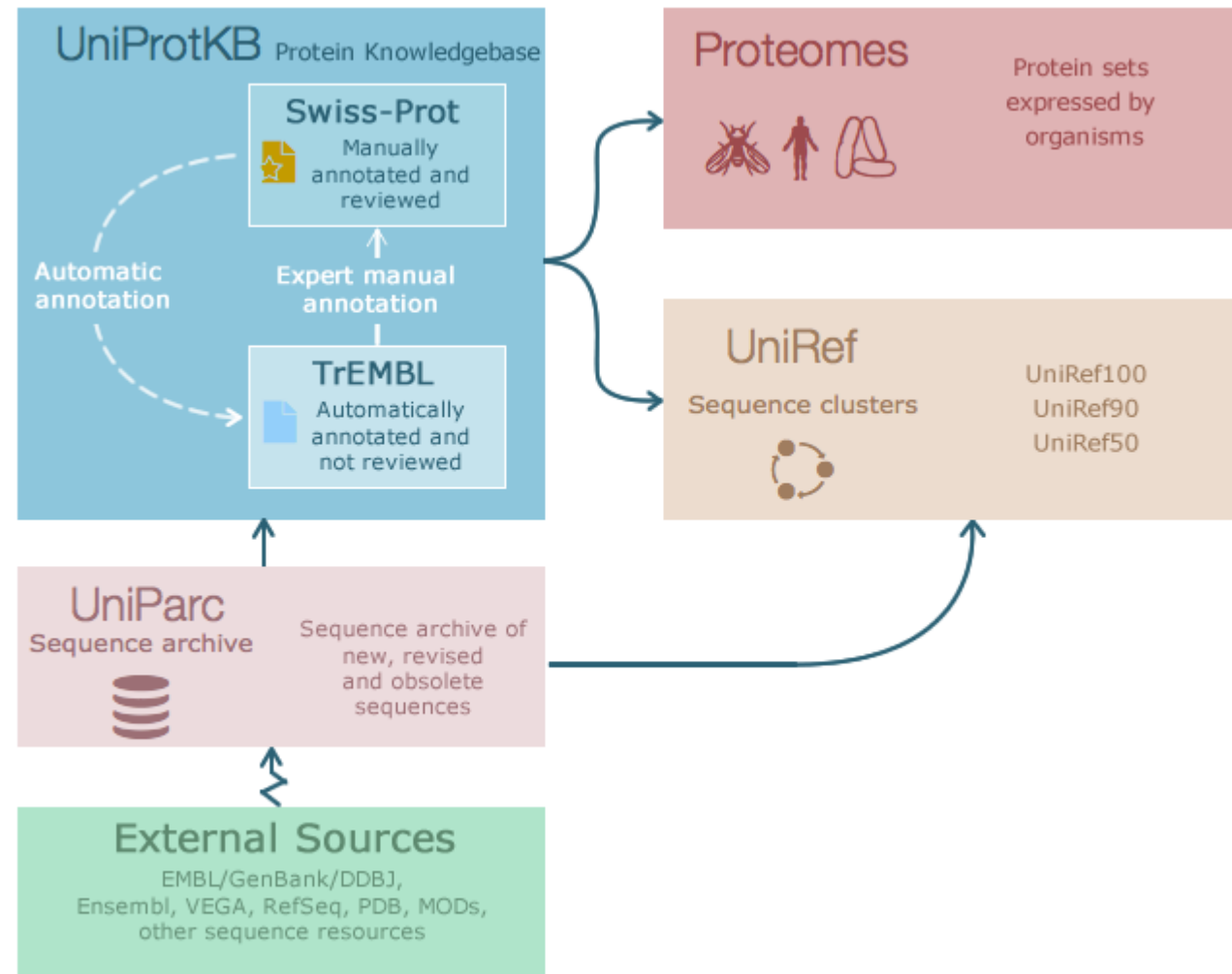
GOLD Biosample

GOLD Project → GOLD Analysis Project

- A comprehensive resource for protein sequence and annotation data
- Collaboration between the European Bioinformatics Institute (EMBL-EBI), the SIB Swiss Institute of Bioinformatics and the Protein Information Resource (PIR)
- EMBL-EBI and SIB together used to produce Swiss-Prot and TrEMBL, while PIR produced the Protein Sequence Database (PIR-PSD)
- Available at: <https://www.uniprot.org/>



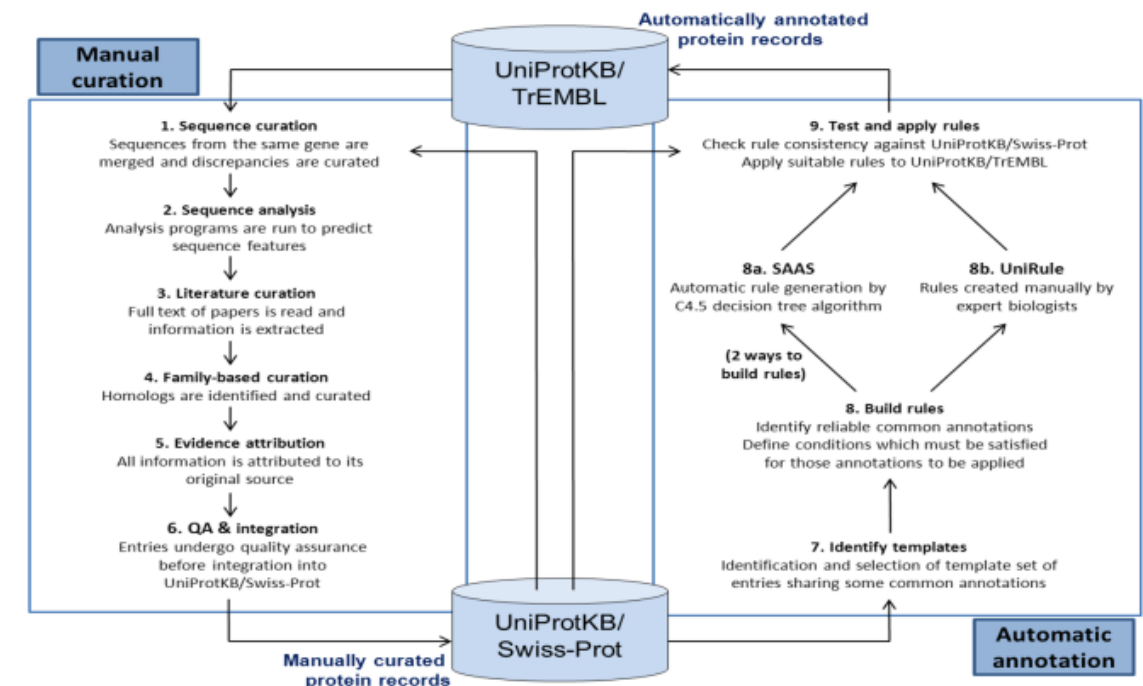
- TrEMBL (Translated EMBL Nucleotide Sequence Data Library) was originally created because sequence data was being generated at a pace that exceeded Swiss-Prot's ability to keep up
- PIR maintained the PIR-PSD and related databases, including iProClass, a database of protein sequences and curated families
- In 2002 the three institutes decided to pool their resources and expertise and formed the UniProt consortium



- More than 95% of the protein sequences provided by UniProtKB come from the translations of coding sequences (CDS) submitted to the ENA/GenBank/DDBJ nucleotide sequence resources
- In addition to translated CDS, UniProtKB protein sequences may come from:
 - ▶ The PDB database
 - ▶ Sequences experimentally obtained by direct protein sequencing and submitted to UniProt
 - ▶ Sequences scanned from the literature
 - ▶ Sequences derived from gene prediction but which have not been submitted to ENA/GenBank/DDBJ, imported from resources such as Ensembl and RefSeq

- Attempts to attach as much functional information as possible to each protein sequence to provide users with an overview of the available information for a given protein
- This information is added
 - manually by the UniProt biocurators who are all trained biologists
 - automatically through various annotation systems which have been developed within the group
- Manual curation consists of a critical review of experimental and predicted data for each protein as well as manual verification of each protein sequence

- Curation methods applied include:
 - ▶ manual extraction and structuring of information from the literature
 - ▶ manual verification of results from computational analyses
 - ▶ mining and integration of large-scale data sets
 - ▶ continuous updating as new information becomes available



Example P05067

- Go to the webpage of the UniProt database (<https://www.uniprot.org/>)
- Search for at least four of the different UniProt identifiers provided in the document ABI_UniProt-Searches
- Please answer all questions for each of the identifiers you picked

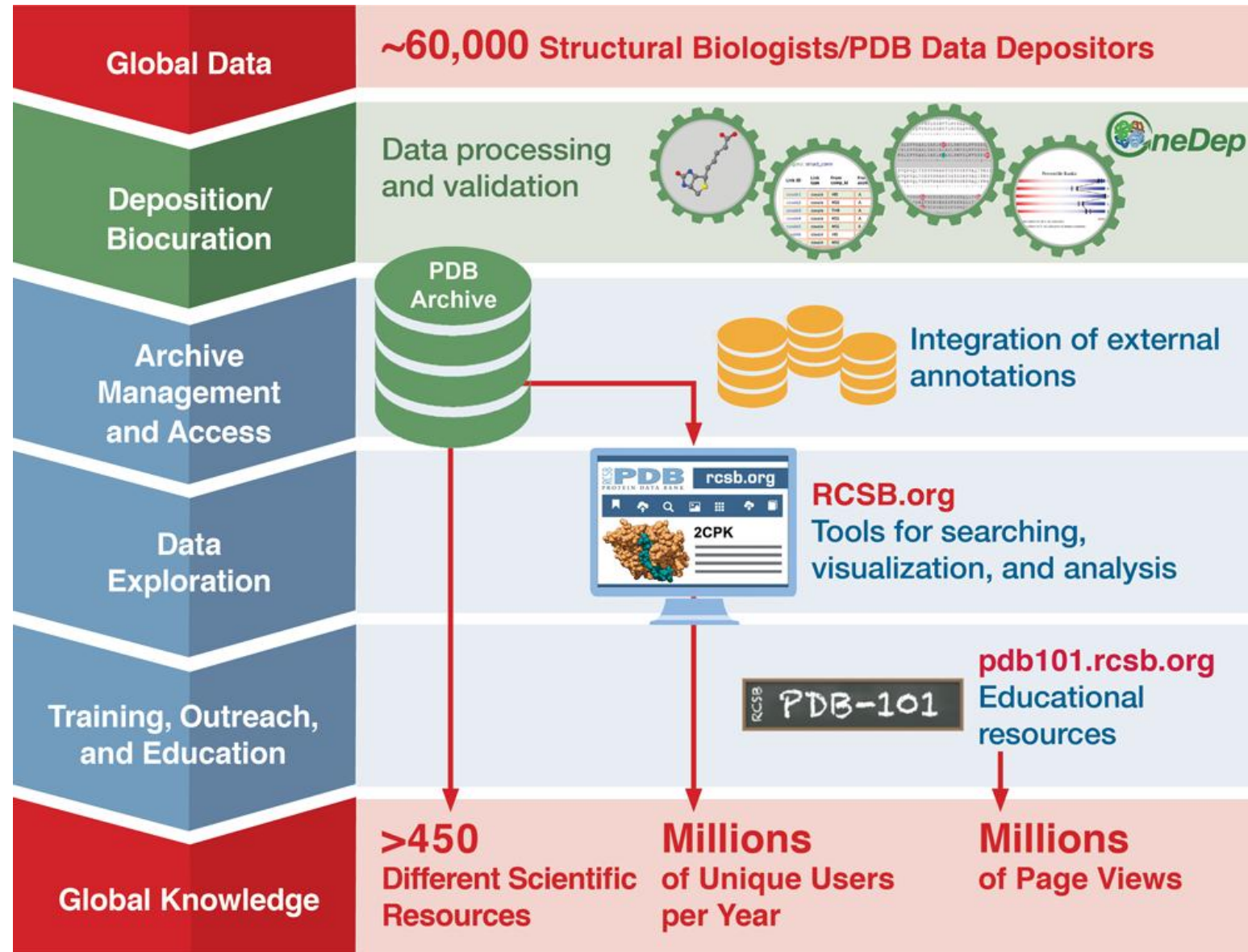
- The US data center for the global Protein Data Bank (PDB) archive of 3D structure data for large biological molecules (proteins, DNA and RNA)
- Established as the 1st open access digital data resource in all of biology and medicine in 1971 as Brookhaven protein data bank
- Provides access to 3D structure data for the molecules of life, found in all organisms on the planet
- Freely online available under <https://www.rcsb.org/>



- Established in 1971 at Brookhaven National Laboratory under the leadership of Walter Hamilton
- Originally contained 7 structures
- Tom Koetzle began to lead the PDB in 1973
- Then Joel Sussman in 1994
- Led by Helen M. Berman, the Research Collaboratory for Structural Bioinformatics (RCSB) became responsible for the management of the PDB in 1998
- In 2003, the Worldwide Protein Data Bank (wwPDB) was formed to maintain a single PDB archive

- The wwPDB consists of 5 members:
 1. RCSB-PDB: the american Protein Data Bank
 2. PDBe: the european Protein Data Bank
 3. PDBj: the japanese Protein Data Bank
 4. BMRB: the Biological Magnetic Resonance Data Bank
 5. EMDB: the Electron Microscopy Data Bank

- Knowing the 3D structure of a biological macromolecule is essential for understanding its role in health and disease, its function in plants, food and energy production, and its importance to other topics
- The enormous wealth of 3D structure data stored in the PDB has underpinned significant advances in our understanding of protein architecture, culminating in recent breakthroughs in protein structure prediction accelerated by artificial intelligence approaches and deep or machine learning methods



- The RCSB-PDB contains around 200000 experimentally-determined structures, that stem from methods like:
 - Cryo-EM
 - Xray Crystallography
 - NMR Methods
- Furthermore it contains around 1000000 computed structures models from:
 - AlphaFold DB
 - ModelArchive

Example 1KIM

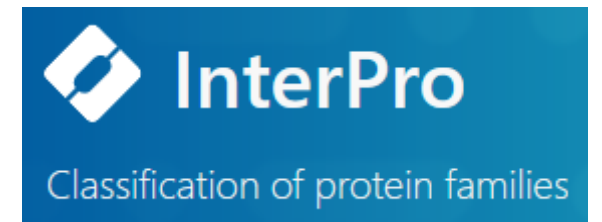
- Go to the webpage of the UniProt database (<https://www.rcsb.org/>)
- Search for at least three of the different PDB identifiers provided in the document ABI_PDB-Searches
- Please answer all questions for each of the identifiers you picked
- Afterwards look at the following structures and write down the methods used for determining it: 1RMN, 1SSZ, 134D, 1C57, 1AT9, 1F6H, 1E07, 1EQ8, 1AGA, 1ZYT
- Use the advanced search of the RCSB-PDB and find out how many structures you can find for each of these methods
- You can find the corresponding filter under Structure Attributes → Methods

Secondary databases

- As mentioned earlier secondary databases contain derived knowledge stemming from experimental data
- They contain information about:
 - ▶ protein-protein interactions (STRING)
 - ▶ protein folds (SCOP)
 - ▶ orthologous groups (eggNOG)
 - ▶ protein classification (InterPro)
 - ▶ Genotype phenotype relations (OMIM)
 - ▶ ...

- Secondary databases collect data from primary databases, compare them and present you the analysis results
- Furthermore they make use of intensive literature research to annotate the data even more
- Often times the analysis leads to the developement of prediction tools
 - These tools take for example a protein sequence and search through for it specific markers related to domains, function, active center etc.

- Provides functional analysis of proteins by classifying them into families and predicting domains and important sites
- To classify proteins in this way, InterPro uses predictive models, known as signatures, provided by several different databases (referred to as member databases) that make up the InterPro consortium
- Combines protein signatures from these member databases into a single searchable resource, capitalising on their individual strengths to produce a powerful integrated database and diagnostic tool
- <https://www.ebi.ac.uk/interpro/>



- This database allows you to search through its 13 member databases with the help of a sequence and thus generating a functional analysis of your protein
- Of course this can be especially useful when working with unknown protein sequences to determine the function and architecture of it
- The combination of 13 different databases with their own specialities provides a wide variety of information for your sequence

- CATH-Gene3D
- CDD
- HAMAP
- PANTHER
- Pfam
- PIRSF
- PRINTS



- PROSITE profiles
- PROSITE patterns
- SFLD
- SMART
- SUPERFAMILY
- TIGRFAMs
- MobiDB



- PHOBIUS: transmembrane topology and signal peptide predictor
- TMHMM: Prediction of transmembrane helices in proteins
- SIGNALP: Signal peptide and cleavage sites
- COILS: Predicting coiled coils from protein sequences

- Simple Modular Architecture Research Tool allows the identification and annotation of protein domains
- Can be used to analyze protein domain architectures
- Contains manually curated models for more than 1300 protein domains
- These domains are extensively annotated with respect to phyletic distributions, functional class, tertiary structures and functionally important residues
- Was never intended to be exhaustive and focused initially on mobile domains, which occur in various contexts while retaining similar function
- The first version contained only 68 signaling domains and focused only on detecting these domains from sequences

- Each domain found in a non-redundant protein database as well as search parameters and taxonomic information are stored in a relational database system
- User interfaces to this database allow searches for proteins containing specific combinations of domains
- For each domain extensive information including cross links and references can be found
- All services are available under <http://smart.embl-heidelberg.de/>

- SMART is able to determine the architectures of single sequences of genomes
- All profiles for the different domains were derived from optimized and analyzed Multiple sequence alignments
- In its original version it was able to detect all annotated domains that were also present in the SwissProt database
- As part of InterPro nowadays it's a good addition to the other available tools like Pfam and PRINTS

- A database of protein families and domains
- Based on the observation that, while there is a huge number of different proteins, most of them can be grouped, on the basis of similarities in their sequences, into a limited number of families
- Proteins or protein domains belonging to a particular family generally share functional attributes and are derived from a common ancestor
- It currently contains patterns and profiles for more than a thousand protein families or domains
- Each of these signatures comes with documentation providing background information on the structure and function of these proteins

- When studying protein sequence families, it's apparent that some regions have been better conserved than others during evolution
- These regions are generally important for the function of a protein and/or for the maintenance of its three- dimensional structure
- Analyzing the constant and variable properties of such groups of similar sequences, allows to derive a signature for a protein family or domain
- A protein signature can be used to assign a newly sequenced protein to a specific family of proteins and thus to formulate hypotheses about its function

Example from Webpage

- Use the UniProt identifiers from the file ABI_UniProt-Searches and use the corresponding sequences for some InterPro Scans
- Scan at least three of these sequences
- Look at the scan results and get accustomed to the different results

- Online Mendelian Inheritance in Man (OMIM®) is a continuously updated catalog of human genes and genetic disorders and traits
- A continuation of Dr. Victor A. McKusick's Mendelian Inheritance in Man, which was published through 12 editions, the last in 1998
- Based on the peer-reviewed biomedical literature
- Criteria for inclusion of papers continue to evolve
- In general, priority for inclusion is given to papers that provide significant insight into the gene-phenotype relationship, expand the understanding of human biology, or contribute to the characterization of a disorder

- Information in each OMIM entry is cited, and the full reference is provided
- Is biocurated at the McKusick-Nathans Institute of Genetic Medicine at the Johns Hopkins University School of Medicine
- Main contents are summaries regarding mutations and corresponding diseases in the human
- Mutations that lead to higher probabilities to develop certain diseases, for example pancreatic cancer are also included

- Currently OMIM contains around 27000 entries (see table below)
- The largest part of this are gene descriptions followed by phenotype descriptions with the known molecular basis

Type	Autosomal	X Linked	Y Linked	Mitochondrial	Total
Gene description	16241	768	51	37	17097
Gene and phenotype	21	0	0	0	21
Phenotype description, basis known	6302	380	5	34	6721
Phenotype description, basis unknown	1394	112	4	0	1510
Other	1640	101	3	0	1744
Totals	25598	1361	63	71	27093

OMIM Mainpage

OMIM Example *608092

OMIM Example #606856

- Develop a comprehensive, computational model of biological systems
- Ranging from the molecular to the organism level
- Across the multiplicity of species in the tree of life

- World's largest source of information on the functions of genes
- Both human-readable and machine-readable
- Foundation for computational analysis of large-scale molecular biology and genetics experiments in biomedical research.

- integrating proteomic information from different organisms
- assigning functions to protein domains
- finding functional similarities in genes that are overexpressed or underexpressed in diseases and as we age
- predicting the likelihood that a particular gene is involved in diseases that haven't yet been mapped to specific genes
- analysing groups of genes that are co-expressed during development
- developing automated ways of deriving information about gene function from the literature
- verifying models of genetic, metabolic and product interaction networks

- Gene Ontology allows you to run so called enrichment analysis with a group of genes
- This analysis can tell you whether certain GO-Terms are over- or under-represented in the uploaded geneset in comparison to a background set
- The result page will show you how many genes of the background set and your set fall under specific categories
- It will also show you how many genes you would expect and some statistical values to help you determine whether the terms are over- or under-represented in your gene set

Gene Ontology Mainpage

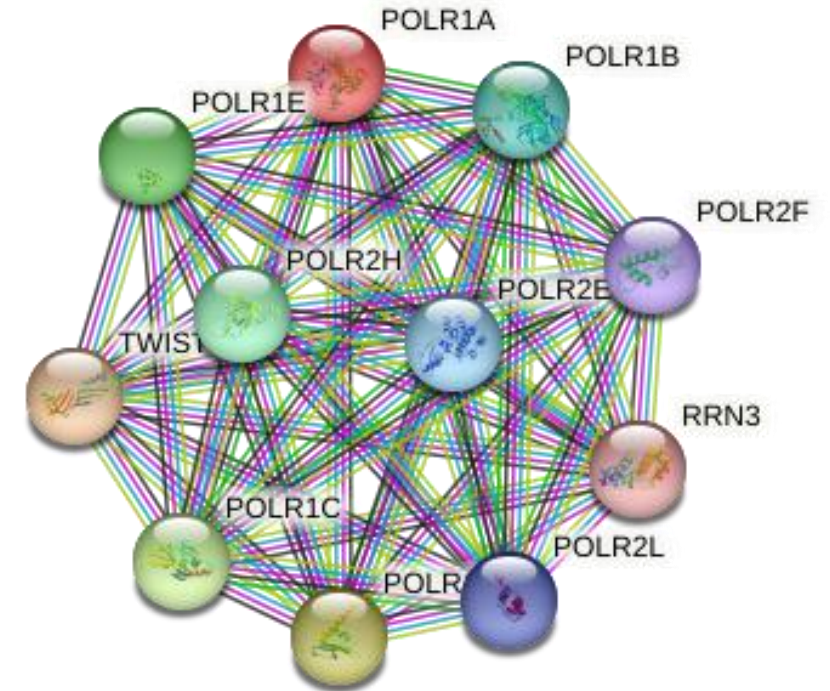
Gene Ontology Example heparin binding

Gene Ontology / PANTHER Example Enrichment

- Database of known and predicted protein-protein interactions
- The interactions include direct (physical) and indirect (functional) associations
- They stem from computational prediction, from knowledge transfer between organisms, and from interactions aggregated from other (primary) databases
- The STRING database currently covers 67'592'464 proteins from 14'094 organisms
- Freely available under <https://string-db.org/>



- The results of STRING are interaction networks as shown on the right
- Each Node (Sphere) represents one protein, with the protein you searched for colored in red (POLR1A)
- The map itself shows a number of different interactions by color coding the lines between the nodes (see next slide)



- The interactions between proteins shown in STRING fall under three categories:
 - ▶ Known interactions
 - light blue lines for interactions derived from curated databases
 - purple lines for experimental evidence
 - ▶ Predicted interactions
 - green lines for gene neighborhood
 - red line for gene fusions
 - blue lines for gene co-occurrence
 - ▶ Others
 - yellow lines for textmining
 - black lines for co-expression

- Genomic Context Predictions
- High-throughput Lab Experiments
- (Conserved) Co-Expression
- Automated Textmining
- Previous Knowledge in other Databases

Examples from Webpage

Database	Description
PHI-base	curated molecular and biological information on genes proven to affect the outcome of pathogen-host interactions http://www.phi-base.org/index.jsp
Ensembl	genome browser for vertebrate genomes annotate genes, computes multiple alignments, predicts regulatory function and collects disease data http://www.ensembl.org/index.html
Ensembl Genomes	Providing genome data for non-vertebrate species, with tools for the manipulation, analysis and visualisation of that data (Plants, Metazoa, Protists, Fungi & Bacteria) http://ensemblgenomes.org

Database	Description
BioGRID	public database that archives and disseminates genetic and protein interaction data from model organisms and humans https://wiki.thebiogrid.org/doku.php/aboutus
IMP	Analyze your experimental results in the functional context of gene-gene networks from multiple organisms http://imp.princeton.edu
Reactome	A curated database of biological processes in humans. It covers biological pathways ranging from the basic processes of metabolism to high-level processes such as hormonal signalling http://reactome.org

- Use the String database to analyze the topics shown in the table below
- For each search note the number of nodes in the network, the number of edges and how many nodes don't have any edges associated with them

Search term / topic	Organisms	Field
wnt-protein binding	homo sapiens	Pathway
endoplasmic reticulum localization	mus musculus	Pathway
manganese ion binding	drosophila melanogaster	Pathway
Amyloid-beta precursor protein	homo sapiens	Protein by Name
Sequence from H2QV72 (UniProt)	Pan troglodytes	Protein by Sequence
COG0265		Protein families („GOGs“)

NCBI-Entrez

- The NCBI was founded in 1988 through legislation sponsored by US Congressman Claude Pepper
- It was originally directed by David Lipman, one of the original authors of the BLAST sequence search tool and a widely respected figure in bioinformatics
- It offers a wide variety of databases and bioinformatic tools



- Molecular database system of the NCBI (National center of biotechnology information) in the US
- Provides access to a multitude of different databases like Gene, Nucleotide, PubMed and so on
- Gives you even access to Structural data of proteins via crosslinks to the PDB
- Can be used to find a nearly all freely available information about a specific biochemical, medical or biotechnological subject

- The databases of the Entrez system can be divided into the following categories
 - ▶ Literature
 - ▶ Genes
 - ▶ Proteins
 - ▶ Genomes
 - ▶ Clinical
 - ▶ Pubchem (only 1 database)
- Each category hosts several databases

- Combines literature for the search topic in the databases
 - ▶ Bookshelf which contains, as the name implies, many books, either physical ones or ebooks
 - ▶ MeSH for medical subject headings, that means medical articles
 - ▶ NLM catalog that provides access to the contents of the National Library of Medicine
 - ▶ PubMed a comprehensive database of citations of scientific articles from different journals
 - ▶ PubMed central an electronic archive of scientific articles

- Gives you information about the genes of a search topic in
 - ▶ Gene an integrated database collection information about the nomenclature, reference sequences and links to resources
 - ▶ GEO Datasets that stores curated gene data sets
 - ▶ GEO Profiles that stores individual gene expressions from the curated datasets of GEO Datasets
 - ▶ PopSet a collection of related DNA sequences derived from population, phylogenetic, mutation and ecosystem studies

- Allows you to access protein data for the search topic in:
 - ▶ Conserved Domains a database for the annotation of functional units in proteins
 - ▶ Identical protein Groups that contains a single entry for each protein translation found in the NCBI resources
 - ▶ Protein a collection of sequences from several sources
 - ▶ Protein Family Models a collection of models representing homologous proteins with a common function
 - ▶ Structure a database for three dimensional protein structures

- You can find information about genomes of a search topic in
 - ▶ Assembly providing information on the structure of assembled genomes
 - ▶ BioCollections a curated dataset of metadata for culture collections, museums, herbaria and other natural history collections
 - ▶ BioProject a collection of biological data related to a single initiative
 - ▶ BioSample contains descriptions of biological source materials used in experimental assays
 - ▶ Genome organizes information on genomes including sequences, maps, chromosomes, assemblies, and annotations

- You can find information about genomes of a search topic in
 - ▶ Nucleotide a collection of sequences from several sources, including GenBank, RefSeq, TPA and PDB
 - ▶ SRA the largest publicly available repository of high throughput sequencing data
 - ▶ Taxonomy a curated classification and nomenclature for all of the organisms in the public sequence databases

- You can find clinical data for a specific topic in
 - ▶ ClinicalTrials a database of privately and publicly funded clinical studies conducted around the world
 - ▶ ClinVar aggregates information about genomic variation
 - ▶ dbGaP the database of Genotypes and Phenotype in humans
 - ▶ dbVar a database of human genomic Structural Variation
 - ▶ GTR a central location for voluntary submission of genetic test information
 - ▶ MedGen organizes information related to human medical genetics
 - ▶ OMIM a comprehensive, authoritative compendium of human genes

- Entrez also gives you access to PubChem
- The world's largest collection of freely accessible chemical information
- Search chemicals by name, molecular formula, structure, and other identifiers
- You can find information about Compounds, pathways, BioAssays, Substances of a specific search topic
- The data displayed on the results page seems to have some problems since it does NOT match the results in the database

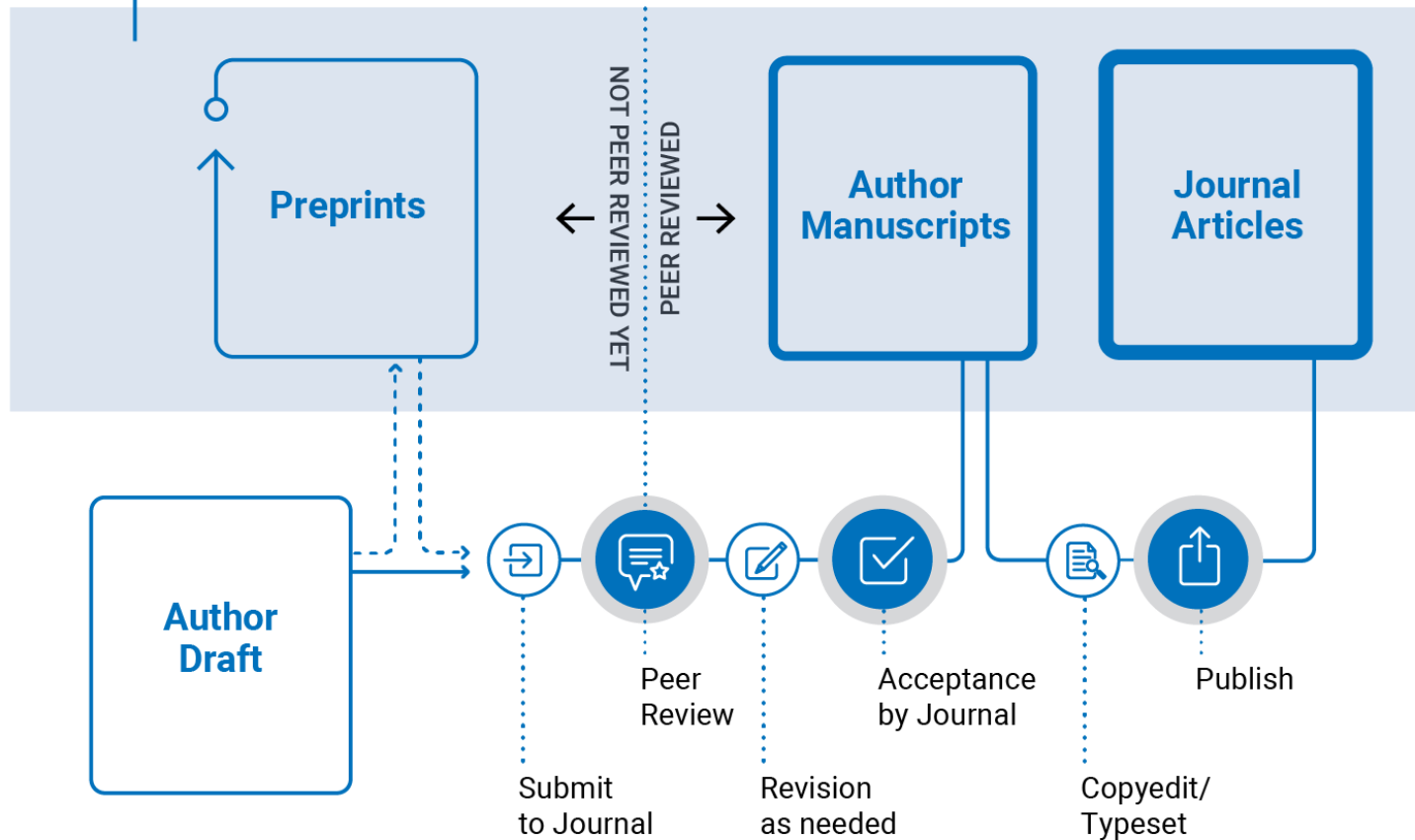
- This database contains protein sequences derived from several sources
- Including all of the following:
 - ▶ Translations from annotated coding regions in GenBank, RefSeq and TPA
 - ▶ Records from SwissProt, PIR, PRF, and RCSB-PDB
- For each sequence saved in the database a wide variety of information (depending on availability) is collected, including:
 - ▶ The organism
 - ▶ References
 - ▶ Structural and/or sequence features
 - ▶ The sequence itself

- This database contains information about many different genes in many different species
- For each entry a long list of information is saved, including:
 - ▶ Genomic context, e.g. the corresponding chromosome
 - ▶ Genomic regions, transcripts and products, e.g. which protein does the gene encode for or where can enhancer / silencer bind
 - ▶ Expression, e.g. in which tissues can it be found
 - ▶ Bibliography, that means a collection of articles in PubMed and other Resources
 - ▶ Phenotypes, e.g. which diseases are linked to this gene
 - ▶ and many more

- A free resource supporting the search and retrieval of biomedical and life sciences literature with the aim of improving health
- The PubMed database contains more than 35 million citations and abstracts of biomedical literature
- It does not include full text journal articles
- However, links to the full text are often present when available from other sources, such as the publisher's website or PubMed Central (PMC)

- A free full-text archive of biomedical and life sciences journal literature at the U.S. National Institutes of Health's National Library of Medicine (NIH/NLM)
- PMC is part of the NLM collection, which also includes NLM's extensive print and licensed electronic journal holdings
- Since its inception in 2000, PMC has grown from comprising only two journals:
 - PNAS: Proceedings of the National Academy of Sciences
 - Molecular Biology of the Cell
- Now it's an archive of articles from thousands of journals
- Also contains author manuscripts and preprints

What's in PubMed Central (PMC)?



- Contains more than 8 million full-text article records
- Spanning several centuries of biomedical and life science research (late 1700s to present)

- This database is a collection of nucleotide sequences (RNA / DNA) stemming from a multitude of different sources:
 - ▶ GenBank
 - ▶ RefSeq
 - ▶ TPA
 - ▶ RCSB-PDB
- The sequences are not annotated and thus it contains a lot of redundancy

- GenBank ® is the NIH genetic sequence database, an annotated collection of all publicly available DNA sequences
- Is part of the INSDC just like DDBJ and ENA
- A GenBank release occurs every two months
- Most of the data stems from direct submissions from many different scientists

- The NCBI combined all genomic resources (Assembly and Genome) under the new NCBI datasets
- This webpage provides a single access point on genomic data
- The data is presented with a focus on taxonomy
- Each genome is presented alongside Assembly information, literature citations and metadata

Example Hepatitis C NS3

- Please use the search topics I uploaded to the Nextcloud and answer the questions
- You don't have to finish all topics since one topic will probably take around 30 to 60 minutes
- Try to finish at least three topics of your choice
- If you still have time after finishing your three topics you can explore the databases on your own afterwards

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